

Mohammad Machaka

machaka.mohammad@gmail.com | +33(0)7 53 37 78 23 | Toulouse, FR | machaka.net
| linkedin.com/in/mohammad-machaka-a63685172

PROFILE

Systems-minded software and cloud engineer transitioning into Systems Engineering through the ISAE-SUPAERO SEN programme. I design and build reliable, real-time data platforms, bridging formal systems engineering (requirements, trade-offs, and lifecycle management) with hands-on technical delivery. I thrive on breaking down complex problems, making clear design choices, and shipping maintainable services for mission-critical domains.

CERTIFICATIONS

ASEP — INCOSE (May 2026)
AWS Solutions Architect — Associate

SKILLS

SYSTEMS ENGINEERING & MBSE

Requirements • ICDs • Trade studies • V-model • ISO/IEC 15288 • V&V • RCA & CoNQ • Capella / SysML

PROGRAMMING

C++ • Python • C# • Java • SQL • TypeScript • Node.js

CLOUD & DEVOPS

AWS • Azure • Kubernetes • Docker • Terraform • Jenkins • CI/CD • IaC

DATA & ARCHITECTURE

Skywise • Airflow • MongoDB • PostGIS • GEE • REST APIs • Microservices

OBSERVABILITY

Prometheus • Grafana • Alertmanager

LANGUAGES

English (Fluent) • French (Advanced) • Arabic (Fluent)

EXPERIENCE

AIRBUS ELECTRIC CENTER | SYSTEMS & QUALITY ENGINEERING

May 2026 -- Present • Toulouse, FR

- Architecting a centralized **Skywise** data pipeline to replace legacy tracking, targeting NC and CoNQ reduction.
- Analyzing historical records to investigate high-volume errors and performing RCA to identify areas for process improvement.
- Proposing data-driven solutions and building dashboards to track KPIs and prioritize future preventive actions.

Stack: Skywise, Data Analysis, RCA, Dashboarding, CoNQ, Python

GREEN PRAXIS | CLOUD & DATA ENGINEER • PERMANENT

Jan 2025 -- Aug 2025 • Aix-en-Provence, FR

- Built 15+ **Airflow** pipelines querying STAC to ingest Sentinel/Landsat data (Rasterio/GDAL), increasing throughput 6x (to 48 scenes/hour).
- Provisioned AWS **EKS** nodes using **Terraform** and **Karpenter**, automating dynamic autoscaling and efficiency.
- Rebuilt tiles to on-demand **FastAPI**+GEE (-80% storage, <30m latency).
- Deployed **Prometheus/Grafana** via Helm, defining 35 alert rules; migrated JWT to PASETO, lowering API P95 latency by 63% to 220ms.

Stack: Terraform, AWS, EKS, Airflow, FastAPI, Node, Python, Docker, MongoDB

AIRBUS SAS | DATA ENGINEER • ALTERNANCE

Jan 2023 -- Jun 2024 • Toulouse, FR

- Designed a unified HR reporting model consolidating 5 regions; built Python ETL flows reducing dashboard deployment time by ~80%.

Stack: C++, Python, PostGIS, Architecture Definition, Dashboarding

MUREX SYSTEMS | SOFTWARE ARCHITECT • PROJECT LEAD

May 2021 -- Jan 2022 • Beirut, LB

- Conceived and led a log-analysis platform with clear error taxonomy, parsing logic, and alerting, identifying 50+ previously uncaught patterns.

Stack: Python, Jenkins, Docker, Terraform, Kubernetes

ZAKA | AI PIPELINE DEVELOPER

Oct 2020 -- September 2021 • Beirut, LB

Stack: C++, Python, Nvidia DeepStream, CUDA, TensorRT, Docker, AWS

URBANSELLER | FREELANCE FULL-STACK DEVELOPER

Jun 2020 -- Feb 2021 • Beirut, LB

Stack: C#, .NET, Angular, Azure, SQL Server, REST APIs

EDUCATION

ISAE-SUPAERO | MASTERE SPECIALISE[®] SYSTEMS ENGINEERING (SEN)

Sep 2025 -- May 2026 • Toulouse, FR

UNIV. TOULOUSE III — PAUL SABATIER | M.Sc. COMPUTER SCIENCE FOR AEROSPACE

Sep 2021 -- Jun 2024 • Toulouse, FR

AMERICAN UNIV. OF BEIRUT | B.Sc. COMPUTER SCIENCE • MINOR BUSINESS ADMINISTRATION

Sep 2017 -- Jun 2020 • Beirut, LB

- LCPC Competitive Programming — 2nd place; Valedictorian 2017/18.